

Bellabeat Case Study: Smart Device Usage Analysis

Introduction

Bellabeat is a successful small health technology company that develops smart wellness products for women. Its products help users track their activity, sleep, stress, menstrual cycle, and overall health. The purpose of this analysis is to use Fitbit fitness tracker data to better understand user behavior. The insights gained from this analysis can help Bellabeat improve its marketing strategy and strengthen its position in the global smart device market.

Business Task

The main business task of this project is to analyze user behavior and understand how users interact with their smart devices. The goal is to identify trends in fitness tracker data and provide insights that can help Bellabeat improve its marketing strategy and strengthen its position in the smart device market.

Data Source

The FitBit Fitness Tracker Data is a public dataset collected from respondents through a distributed survey and made available on Kaggle. The dataset contains activity and sleep records from 33 users. In this project, two datasets were used: `dailyActivity_merged.csv` and `sleepDay_merged.csv`. The data covers approximately one month, from April to May 2016, providing information about users' daily activity and sleep patterns.

Data Cleaning

The data cleaning process included checking for duplicate records and missing values. Duplicate records were found and removed from the `sleepDay_merged.csv` dataset, while no missing values were detected in the datasets used for the analysis. The date columns were converted to the datetime format to ensure consistency. After cleaning, the activity and sleep

datasets were merged using Id and Date. Finally, a new column called ActivityLevel was created to classify users based on their average daily steps.

Exploratory Data Analysis (EDA)

Analysis 1: Distribution of Daily Steps

We analyzed the distribution of daily steps using a histogram to understand how users' daily activity is distributed. This analysis helps identify common activity patterns, the range of daily steps, and whether most users achieve recommended daily activity levels.

Observation

- Most daily step counts fall between 4,000 and 12,000 steps.
- A small number of days exceed 18,000 steps.
- The distribution is right-skewed due to a few days with exceptionally high step counts.

Business Insight

Bellabeat should encourage users who take fewer than 10,000 steps per day to achieve their daily step goals by offering personalized reminders, activity challenges, and motivational rewards. This could help increase user engagement and promote healthier lifestyles.

Analysis 2: Correlation between Daily Steps and Calories

We analyzed the relationship between daily steps and calories burned using a scatter plot. This analysis was conducted to determine whether users who take more steps tend to burn more calories and to better understand the association between physical activity and calorie expenditure.

Observation

- The scatter plot shows an upward trend.
- There is a moderate positive relationship between daily steps and calories burned.
- A few data points are far from the main cluster, indicating variability in users' activity levels.

Business Insight

Bellabeat should encourage users to achieve their daily step goals by offering personalized reminders, activity challenges, and motivational rewards. This could help users burn more calories, maintain a more active lifestyle, and improve their overall health.

Analysis 3: Correlation between Sleep Duration and Daily Steps

We analyzed the relationship between sleep duration and daily steps using a scatter plot. This analysis was conducted to determine whether sleep duration affects daily physical activity and to better understand the relationship between sleep duration and daily steps.

Observation

- The scatter plot shows a slight downward trend.
- There is a weak negative relationship between sleep duration and daily steps.
- A few data points are far from the main cluster, indicating variability in users' activity levels.

Business Insight

Although the dataset does not show a significant relationship between sleep duration and daily steps, Bellabeat should continue encouraging healthy sleep habits because adequate sleep is essential for overall health and well-being. Since sleep duration alone does not explain physical activity levels, Bellabeat may benefit from analyzing additional factors such as sedentary time, calories burned, or activity intensity to gain deeper insights into user behavior.

Analysis 4: User Activity Segmentation

We classified users into activity levels based on their average daily steps and visualized the results using a bar chart. This analysis was conducted to understand the distribution of users across different activity levels and identify which groups represent the largest proportion of users.

Observation

- The largest user groups were Low Active and Somewhat Active, each representing 27.3% of users (9 users).
- Sedentary users represented 24.2% of users (8 users).
- Only seven users were classified as Active (21.2%).

Business Insight

Bellabeat should focus on encouraging users who are classified as Low Active or Somewhat Active to achieve their daily step goals by offering personalized reminders, activity challenges, and motivational rewards. This strategy could increase user engagement and promote healthier lifestyles.

Analysis 5: Average Sleep Duration by Activity Level

We merged users' activity levels with their average sleep duration and visualized the results using a bar chart. This analysis was conducted to compare average sleep duration across different activity levels and determine whether more active users tend to sleep longer.

Observation

- The Low Active group had the highest average sleep duration.
- The Active group had the lowest average sleep duration.
- There is no clear trend showing that more active users consistently sleep longer.

Business Insight

Bellabeat should continue promoting healthy sleep habits because adequate sleep is important for overall health and well-being. Since the data does not show a strong relationship between sleep duration and activity level, the company should provide personalized recommendations for sleep and physical activity separately rather than assuming that improving one will automatically improve the other.

Key Findings

1. Moderate positive relationship between daily steps and calories burned.

Users who take more daily steps generally burn more calories. However, calorie expenditure is also influenced by other factors such as age, nutrition, and lifestyle.

2. Weak relationship between sleep duration and daily activity.

The analysis found only a weak negative relationship between sleep duration and daily steps, indicating that longer sleep does not necessarily lead to higher physical activity.

3. Most users are not highly active.

The majority of users were classified as Low Active or Somewhat Active, suggesting that many users have room to improve their daily physical activity.

4. Activity level does not clearly determine sleep duration.

Comparing average sleep duration across activity levels showed no consistent trend that more active users sleep longer.

Recommendations

1. Encourage inactive users

Bellabeat should encourage Low Active and Somewhat Active users through personalized step goals, reminders, and activity challenges.

2. Promote healthy lifestyles

The Bellabeat app should motivate users to increase their daily physical activity because higher activity is associated with greater calorie expenditure.

3. Continue promoting healthy sleep habits

Although the analysis did not find a strong relationship between sleep duration and activity level, healthy sleep remains important for overall well-being.

4. Provide personalized recommendations

Bellabeat should provide separate recommendations for sleep and physical activity instead of assuming improvements in one will automatically improve the other.

Limitations

- The dataset includes data from only **33 users**, which may not represent the entire population.
 - The dataset covers only **one month** of user activity.
 - Calories burned are influenced by additional factors such as age, weight, nutrition, and metabolism, which were not included in the dataset.
 - Sleep quality, stress level, and health conditions were not available for analysis.
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Conclusion

This analysis provided valuable insights into users' activity and sleep behaviors using Fitbit data. The results showed a moderate positive relationship between daily steps and calories burned, while sleep duration had only a weak relationship with physical activity. Additionally, most users were classified as Low Active or Somewhat Active, highlighting an opportunity for Bellabeat to increase user engagement through personalized recommendations, activity challenges, and health-focused features. These findings can help Bellabeat improve its marketing strategy and support its goal of becoming a stronger competitor in the global smart device market.

References

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